

Continuous exposures with propensity scores

Malcolm Barrett

Stanford University

Warning! Propensity
score weights are
sensitive to positivity
violations for
continuous exposures.

The story so far

Propensity score weighting

- 1 Fit a propensity model predicting exposure x , $x + z$ where z is all covariates
- 2 Calculate weights
- 3 Fit an outcome model estimating the effect of x on y weighted by the propensity score

Continuous exposures

- 1 Use a model like `lm(x ~ z)` for the propensity score model.
- 2 Use `wt_ate()` with `.fitted` and `.sigma`; transforms using `dnorm()` to get on probability-like scale.
- 3 Apply the weights to the outcome model as normal!

Alternative: quantile binning

- 1 Bin the continuous exposure into quantiles and use categorical regression like a multinomial model to calculate probabilities.
- 2 Calculate the weights where the propensity score is the probability you fall into the quantile you actually fell into. Same as the binary ATE!
- 3 Same workflow for the outcome model

1. Fit a model for exposure ~ confounders

```
1 model <- lm(  
2   exposure ~ confounder_1 + confounder_2,  
3   data = df  
4 )
```

2. Calculate the weights with `wt_ate()`

```
1 model |>
2   augment(data = df) |>
3   mutate(wts = wt_ate(
4     .fitted,
5     exposure,
6     # .sigma is from augment()
7     .sigma = mean(.sigma, na.rm = TRUE),
8     exposure_type = "continuous"
9   ))
```

Does change in smoking intensity (**smkintensity82_71**) affect weight gain among lighter smokers?

```
1 nhfs_light_smokers <- nhfs_complete |>  
2   filter(smokeintensity <= 25)
```

1. Fit a model for exposure ~ confounders

```
1 nhfs_model <- lm(  
2   smkintensity82_71 ~ sex + race + age + I(age^2) +  
3   education + smokeintensity + I(smokeintensity^2) +  
4   smokeyrs + I(smokeyrs^2) + exercise + active +  
5   wt71 + I(wt71^2),  
6   data = nhfs_light_smokers  
7 )
```

2. Calculate the weights with `wt_ate()`

```
1 nhefs_wts <- nhefs_model |>
2   augment(data = nhefs_light_smokers) |>
3   mutate(wts = wt_ate(
4     .fitted,
5     smkintensity82_71,
6     .sigma = mean(.sigma, na.rm = TRUE),
7     exposure_type = "continuous"
8   ))
```

2. Calculate the weights with `wt_ate()`

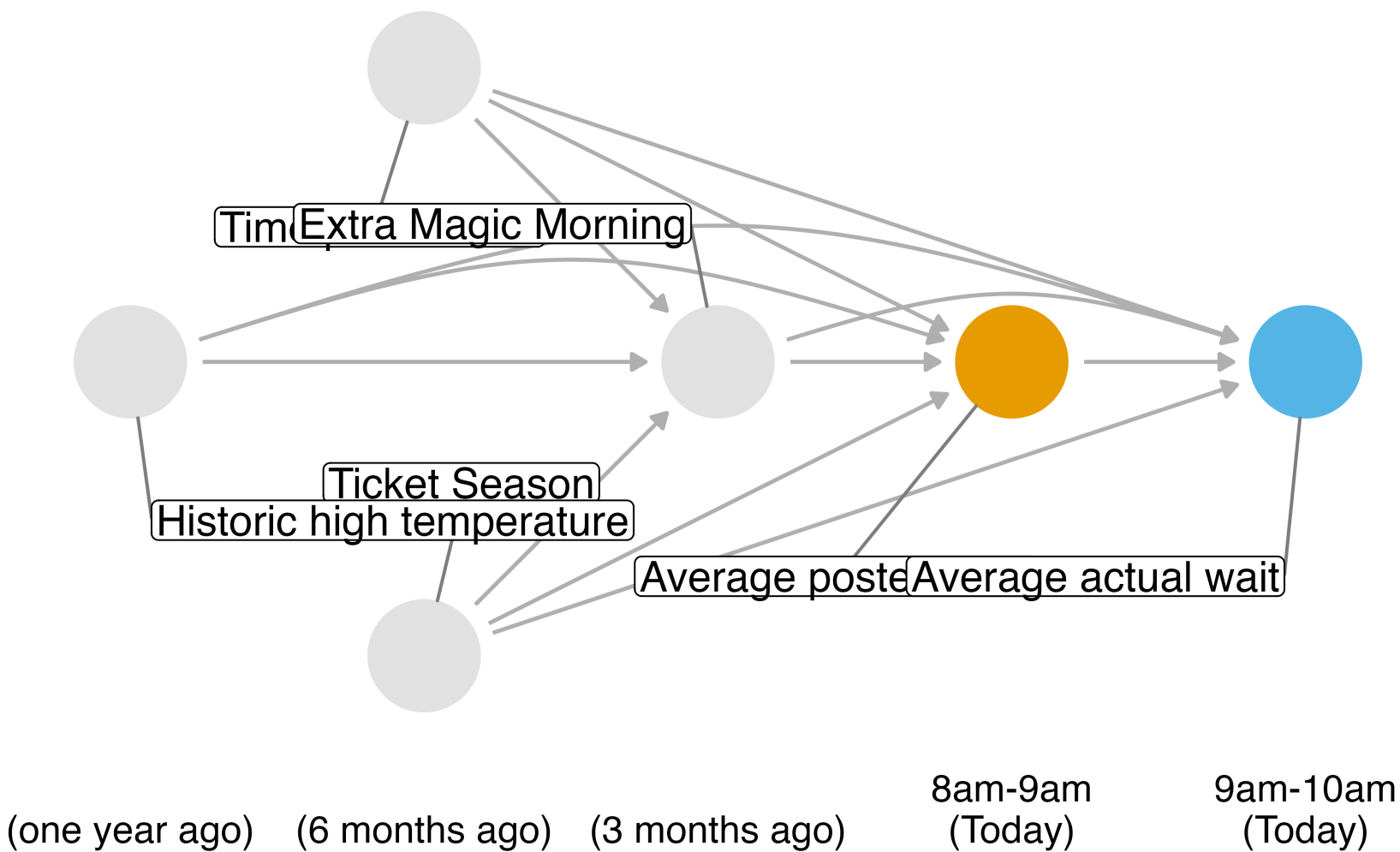
```
1 nhefs_wts
```

```
# A tibble: 1,162 × 74
```

	seqn	qsmk	death	yrdth	modth	dadth	sbp	dbp	sex
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>
1	235	0	0	NA	NA	NA	123	80	0
2	244	0	0	NA	NA	NA	115	75	1
3	245	0	1	85	2	14	148	78	0
4	252	0	0	NA	NA	NA	118	77	0
5	257	0	0	NA	NA	NA	141	83	1
6	262	0	0	NA	NA	NA	132	69	1
7	266	0	0	NA	NA	NA	100	53	1
8	419	0	1	84	10	13	163	79	0
9	420	0	1	86	10	17	184	106	0
10	434	0	0	NA	NA	NA	127	80	1

```
# ... 1,152 more rows
```

Do *posted* wait times at 8 am affect *actual* wait times at 9 am?



Your Turn 1

Fit a model using `lm()` with `wait_minutes_posted_avg` as the outcome and the confounders identified in the DAG.

Use `augment()` to add model predictions to the data frame

In `wt_ate()`, calculate the weights using `.fitted`, `wait_minutes_posted_avg`, and the mean of `.sigma`

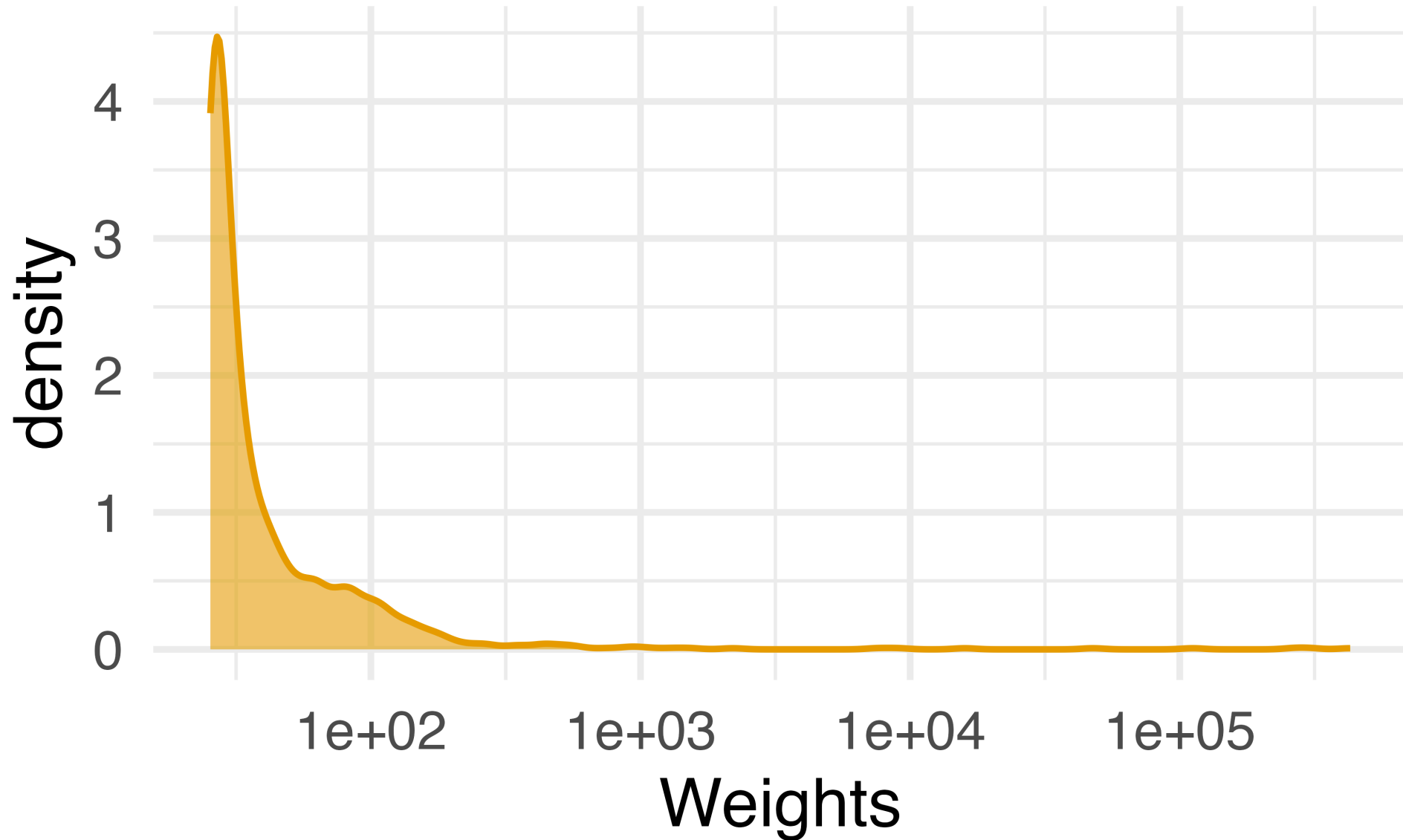
Your Turn 1

```
1 post_time_model <- lm(  
2   wait_minutes_posted_avg ~  
3     park_close + park_extra_magic_morning +  
4     park_temperature_high + park_ticket_season,  
5   data = wait_times  
6 )
```

Your Turn 1

```
1 wait_times_wts <- post_time_model |>
2   augment(data = wait_times) |>
3   mutate(wts = wt_ate(
4     .fitted,
5     wait_minutes_posted_avg,
6     .sigma = mean(.sigma, na.rm = TRUE),
7     exposure_type = "continuous"
8   ))
```

Stabilizing extreme weights



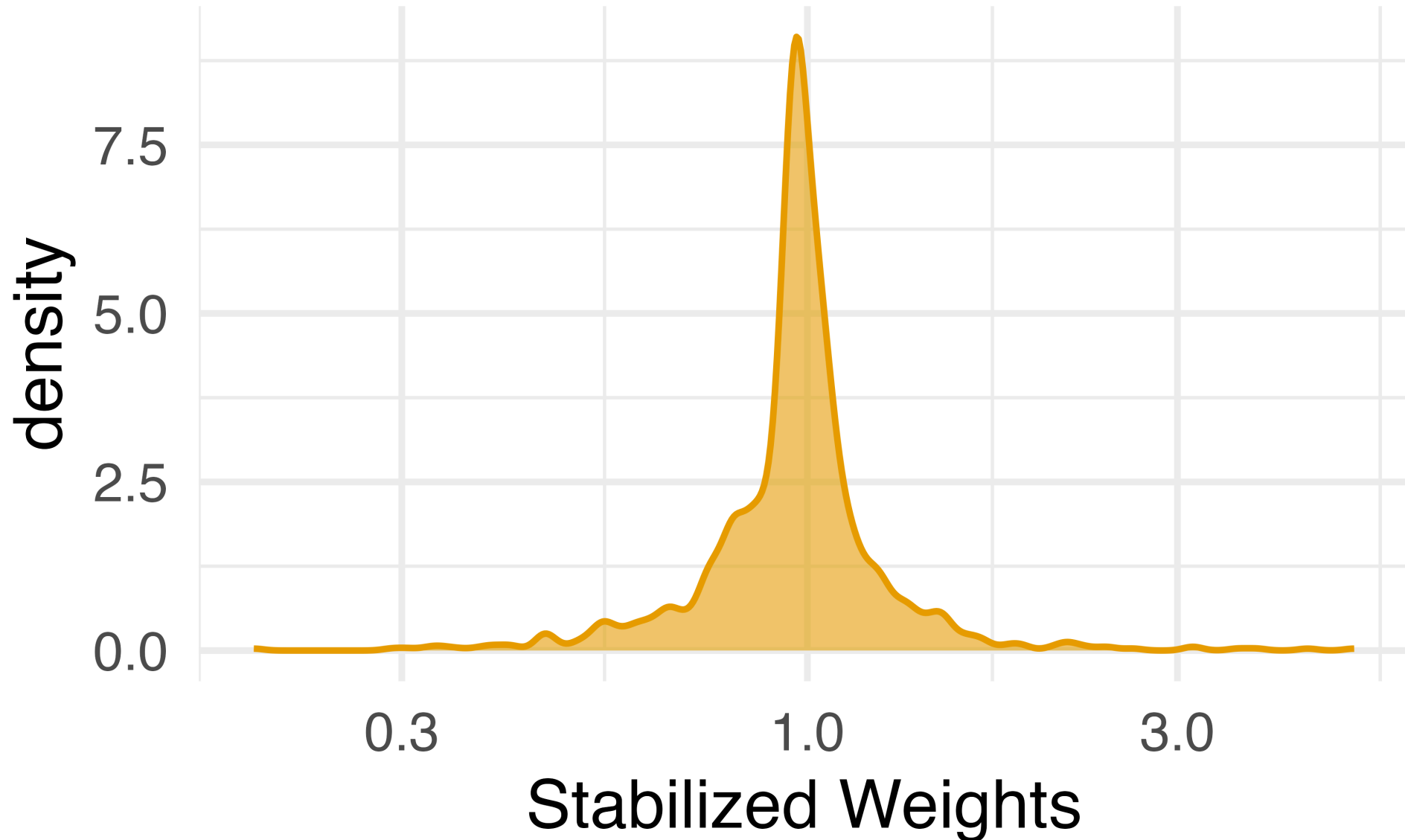
Stabilizing extreme weights

- 1 Fit an intercept-only model (e.g. `lm(x ~ 1)`) or use mean and SD of `x`
- 2 Calculate the marginal density from this model.
- 3 Divide the marginal density by the propensity score density. `wt_ate(..., stabilize = TRUE)` does this all!

Calculate stabilized weights

```
1 nhefs_swts <- nhefs_model |>
2   augment(data = nhefs_light_smokers) |>
3   mutate(swts = wt_ate(
4     .fitted,
5     smkintensity82_71,
6     .sigma = mean(.sigma, na.rm = TRUE),
7     exposure_type = "continuous",
8     stabilize = TRUE
9   ))
```

Stabilizing extreme weights



Stabilized weights should have a mean close to 1

```
1 mean(nhefs_swts$swts)
```

```
[1] 0.9967652
```

If the mean is far from 1, we may have issues with model misspecification or positivity violations. A check worth running every time.

Your Turn 2

Re-fit the above using stabilized weights

Your Turn 2

```
1 wait_times_swts <- post_time_model |>
2   augment(data = wait_times) |>
3   mutate(swts = wt_ate(
4     .fitted,
5     wait_minutes_posted_avg,
6     .sigma = mean(.sigma, na.rm = TRUE),
7     exposure_type = "continuous",
8     stabilize = TRUE
9   ))
```

Fitting the outcome model

- 1 Use the stabilized weights in the outcome model. Nothing new here!

```

1 lm(
2   wt82_71 ~ smkintensity82_71,
3   weights = swts,
4   data = nhfs_swts
5 ) |>
6 tidy() |>
7 filter(term == "smkintensity82_71") |>
8 mutate(estimate = estimate * -10)

```

A tibble: 1 × 5

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	smkintensity82_71	0.960	0.0210	-4.57	0.000000527

Your Turn 3

Estimate the relationship between posted wait times and actual wait times using the stabilized weights we just created.

Your Turn 3

```
1 lm(  
2   wait_minutes_actual_avg ~ wait_minutes_posted_avg,  
3   weights = swts,  
4   data = wait_times_swts  
5 ) |>  
6 tidy() |>  
7 filter(term == "wait_minutes_posted_avg") |>  
8 mutate(estimate = estimate * 10)
```

A tibble: 1 × 5

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	wait_minutes_posted_...	-2.64	0.0803	-3.28	0.00151

Causal dose-response functions

- 1 Calculate the stabilized weights
- 2 Fit the outcome model with a flexible term for the exposure
- 3 Use `ipw()` for standard errors that account for the weights
- 4 Predict the outcome across the range of the exposure

1. Calculate the stabilized weights

```
1 nhefs_dose_wts <- nhefs_light_smokers |>
2   mutate(swts = wt_ate(
3     fitted(nhefs_model),
4     smkintensity82_71,
5     # ipw() rebuilds these weights from the model,
6     # so wt_ate() estimates the SD instead of .sigma
7     exposure_type = "continuous",
8     stabilize = TRUE
9   ))
```

2. Fit a flexible outcome model

```
1 nhefs_msm <- lm(  
2   wt82_71 ~ splines::ns(smkintensity82_71, df = 3),  
3   weights = swts,  
4   data = nhefs_dose_wts  
5 )
```

3. Estimate the effects with `ipw()`

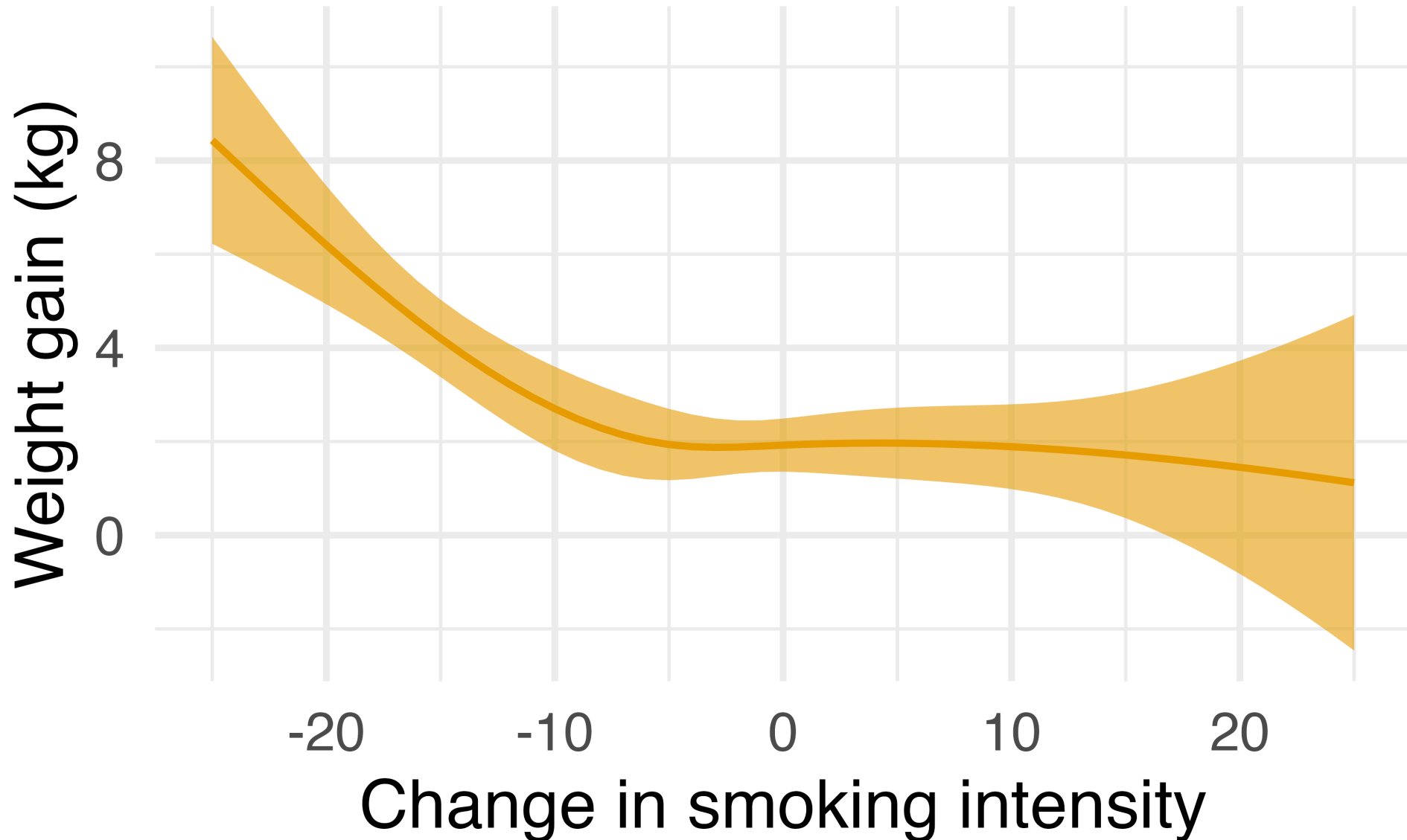
```
1 nhefs_ipw <- ipw(  
2   nhefs_model,  
3   nhefs_msm,  
4   .data = nhefs_dose_wts  
5 )
```

The standard errors account for the estimated weights.

4. Predict across the exposure range

```
1 dose_response <- nhefs_ipw |>
2   as_conditional() |>
3   # marginalesffects needs the data to build the grid
4   set_modeldata(nhefs_dose_wts) |>
5   predictions(
6     newdata = datagrid(smkinsensity82_71 = -25:25)
7   )
```

The causal dose-response function



Diagnosing issues

- 1 Extreme weights even after stabilization
- 2 Bootstrap: non-normal distribution
- 3 Bootstrap: estimate different from original model

More info

Causal Inference in R: g-computation for continuous exposures

<https://github.com/LucyMcGowan/writing-positivity-continuous-ps>